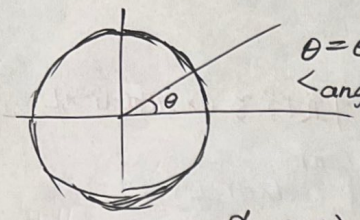
Rotational motion of rigid bodies

角位置：angular position
角速度：angular velocity
角加速度：angular acceleration



$\theta = \theta(t)$
<angular position

do not mix up

$\omega_z = \omega_z(t) = \frac{d}{dt}\theta(t)$
<angular velocity

四指朝旋转方向握住，大拇指即为 ω_z 的方向.

$\alpha_z = \alpha_z(t) = \frac{d}{dt}\omega_z(t) = \frac{d^2}{dt^2}$
<angular acceleration

角加速度 α 的方向由角速度的变化率决定

Rotational Kinematics:

$\omega_f = \omega_i + \alpha t$

$\Delta\theta = \omega_i t + \frac{1}{2}\alpha t^2$

$\Delta\theta = \frac{1}{2}(\omega_i + \omega_f)t$

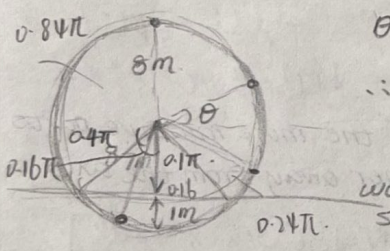
$\omega_f^2 - \omega_i^2 = 2\alpha\Delta\theta$

Problem 1

(i) $n = 6$ rev/min

$\omega = 6\times 2\pi / 60 = 0.2\pi$ rad/s $T = 10$ s

$\theta(3) = \omega \times 3 = 0.6\pi < \pi$



$\theta_i = -0.1\pi$

$\therefore \theta_t = -0.1\pi + 0.2\pi \times 17.5$
$= 3.4\pi$

$3.4\pi - 3\pi = 0.4\pi$

$8\times\cos(0.1\pi) = 7.6084$ $\therefore$ 0.61 m below

这里计算器的角度单位是度，无°也会 by default
必须写 cos(18)

(ii) $h = 7 + 8\sin\theta$

$= 7 + 8\sin(0.2\pi t - 0.1\pi) = 0$

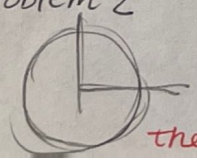
$\therefore \sin(0.2\pi t - 0.1\pi) = -\frac{7}{8}$

$0.2\pi t - 0.1\pi = \arcsin(-\frac{7}{8}) = 0.339\pi$

$t = -1.196+$

$1.76\pi / 0.2\pi = 8.8$ s $\quad 8.8 + \frac{1.68\pi}{0.2\pi} = 17.2$ s

Problem 2



$27.5 - 0.3\times 10 = 24.5$ m/s

$27.5\times 0.3 - \frac{1}{2}\cdot 10\cdot 0.09 = 7.8$ rad 无π!

the line PQ makes an angle $(7.8 - 2\pi) = 1.52$ rad with x axis

$\theta = \frac{s}{r}$ rad, So $s = r\theta$

Linear speed

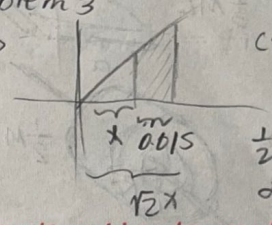
$v = \left|\frac{ds}{dt}\right| = \left|\frac{r\,d(\theta)}{dt}\right| = r\omega \quad \omega = \frac{v}{r}$

$a_t = r\frac{d^2\theta}{dt^2} = r\frac{d\omega_z}{dt} = r\alpha_z \quad a = r\alpha$

$a_c = -\frac{v^2}{r} = -r\omega^2 \quad |a_c| = \omega^2 r$

Problem 3

(i)



$(\sqrt{2}-1)x = 0.61$
$x = 1.47$ s (ii)

$\frac{1}{2}\alpha t^2 = 2\pi$

$\alpha = 1.84\pi$ rad/s^2.

it took 0.61s to make its second complete rev.
<指的是第二rev 0.61s 而不是2rev共0.61s

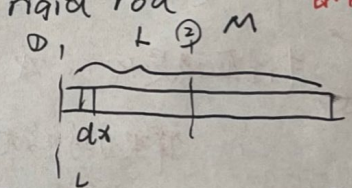
Rotational kinetic energy and moment of inertia

$\Rightarrow KE = \sum \frac{1}{2}mr^2\omega^2 = \frac{1}{2}I\omega^2$

$\Rightarrow I = \sum mr^2 / \int r^2 dm$

I is obtained by multiplying the m of each particle by the square of its ⊥ distance from the axis of rotation and Σ.

rigid rod 看清转轴是哪

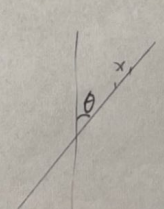


① $\int_0^L \frac{dx}{L}\cdot Mx^2 = \frac{M}{L}\frac{L^3}{3} = \frac{ML^2}{3}$

② $2\int_0^{\frac{L}{2}} \frac{dx}{L}Mx^2 = \frac{2M}{L}\left[\frac{x^3}{3}\right]_0^{\frac{L}{2}} = \frac{ML^2}{12}$

$= \int_{-\frac{L}{2}}^{\frac{L}{2}} \frac{dx}{L}Mx^2$

Problem 4



$2\int_0^{\frac{L}{2}} \frac{dx}{L}Mx^2\sin^2\theta$

$= 2\sin^2\theta\frac{M}{L}\left[\frac{x^3}{3}\right]_0^{\frac{L}{2}} = \frac{ML^2}{12}\sin^2\theta$

平行轴定理

$\boxed{I = I_{cm} + Md^2}$ 平行轴之间的dist

质心在原点，新轴 $x = d$

$I = \int (x+d)^2 dm$

$= \underbrace{\int x^2 dm}_{I_{cm}} + 2d\underbrace{\int x\,dm}_{M} + \underbrace{d^2\int dm}_{Md^2}$

质心pos = 0.

但成刚体的质点到axis的⊥d越大，I越大

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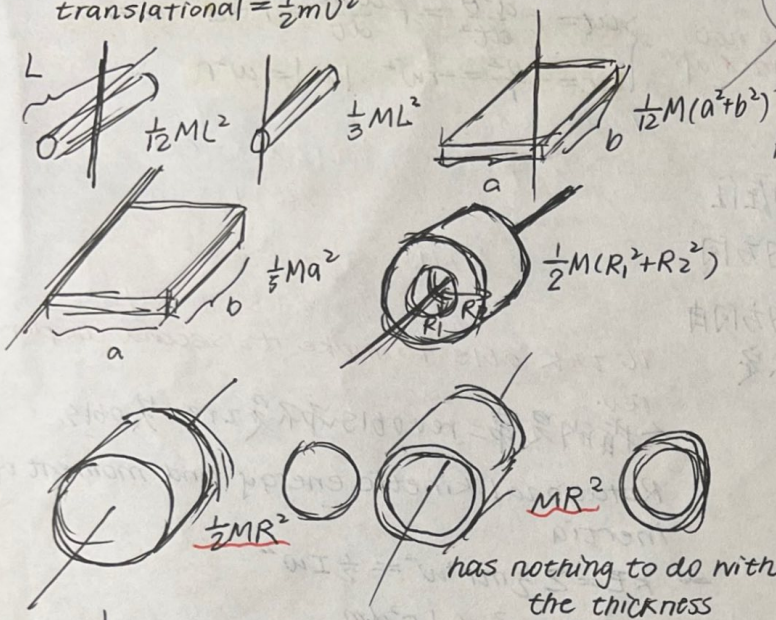
Summary

moment of inertia: $\int r^2 dm$

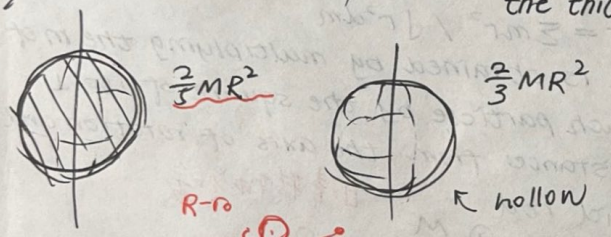
parellel axis theorem: $I = I_{cm} + Md^2$

KE rotational $= \frac{1}{2} I\omega^2$

translational $= \frac{1}{2} mv^2$

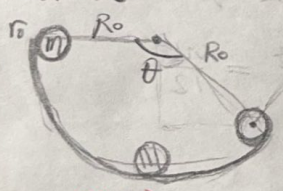


has nothing to do with the thickness



一定注意之后做题 $R-r_0$ 半径应该是 $R-r_0$ ☆

Problem 5



$I = \frac{2}{5} m r_0^2$

算重力势能尤其要注意

下落dist不是R_0了…

$(R-r_0)$

(i) $mgr_0 = \frac{1}{2} m\omega^2 r_0^2 + \frac{1}{2}\frac{2}{5} m r_0^2 \omega^2$

$= \frac{7}{10} m\omega^2 r_0^2 = \frac{7}{10} v^2$

$\therefore v = \sqrt{\frac{10}{7} g (R_0 - r_0)}$

(ii) it descends $(R_0 - r_0)\cos(\theta - \frac{\pi}{2})$

$mg(R_0 - r_0)\cos(\theta - \frac{\pi}{2}) = \frac{7}{10} v^2$

$\therefore v = \sqrt{\frac{10}{7} g (R_0 - r_0)\sin\theta}$

(iii)

$v = (\frac{10}{7} \times 9.8 (0.26 - 0.015) \sin 45°)^{1/2}$

$= 1.56 \, m/s$

$v \times \frac{\sqrt{2}}{2} t - \frac{1}{2} g t^2 = -(R_0 - (R_0 - r_0)\cos 45° - r_0)$

$t = 0.29 s$

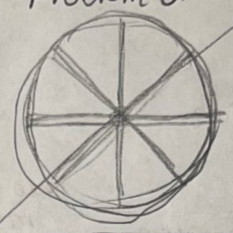
$1.56 \times \frac{\sqrt{2}}{2} \times 0.29 = 0.31 m$

$+ = 0.48$ 看清题…

$(R_0 - r_0)\sin 45° = 0.1 m$

(iv) nearer.

Problem 6



0.6m

$4 \cdot \frac{1}{12} 0.3 \cdot 0.6^2$

$I = 8 \cdot \frac{1}{3} 0.3 \cdot 0.3^2 + 1.48 \cdot 0.3^2$

$= 0.21 kg\, m^2$

Problem 7.

可直接写 $\rho \times 2\pi r dr \times L$

$\int_{R_1}^{R_2} \int_0^{2\pi r} dx dr \cdot L \cdot \rho \cdot r^2$

$= \int_{R_1}^{R_2} \rho L r^2 dr \, 2\pi r$

$= 2\pi \rho L \int_{R_1}^{R_2} r^3 dr$

$= 2\pi \rho L \frac{R_2^4 - R_1^4}{4 \cdot 2}$

$= \frac{1}{2}(R_1^2 + R_2^2) \cdot \rho \pi (R_2^2 - R_1^2)$

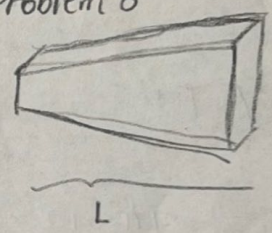
$= \frac{1}{2} m (R_1^2 + R_2^2)$

for solid cylinder, $R_1 \to 0$, $I = \frac{1}{2} m R_2^2$

for thin cylindrical film 薄层, $R_1 \to R_2 = R$, $I = m R_2^2$

Problem 8

(i)



$\frac{dm}{dx} = \beta x$

$dm = \beta x dx$

$m = \int_0^L \beta x dx$

$= \beta \frac{L^2}{2} = \frac{1}{2} \beta L^2$

(ii) $\int_0^L x^2 \beta x dx = \beta \cdot \frac{L^4}{4} = \frac{1}{4} \beta L^4 = \frac{1}{2} m L^2 > \frac{1}{3} m L^2$

I is greater since the more massive parts of the rod are further away from the axis

(iii) $\int_0^L (L-x)^2 \beta x dx = \beta [\int_0^L (L^2 - 2Lx + x^2) x dx]$

$= \beta [L^2 \frac{x^2}{2} - 2L \frac{x^3}{3} + \frac{x^4}{4}]_0^L$

$= \beta [\frac{L^4}{2} - \frac{2}{3} L^4 + \frac{1}{4} L^4] = \beta \frac{6 - 8 + 3}{12} L^4 = \frac{1}{12} \beta L^4 = \frac{1}{6} m L^2$

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Dynamics of rotational motion.

Angular momentum & torque on a particle.

角动量 力矩

$L = I\omega = (mr^2)\omega = (mr^2)(\frac{v}{r})$ (针对于质点)

$\therefore L = rmv = \vec{r}\times\vec{p}$ 角动量是矢量(指向是由r径刚体<180°的角转) > angular momentum $L = I\omega = \vec{r}\times\vec{p}$

rmv — 质点

$\tau = \vec{r}\times\vec{F} = \|r\|\|F\|\sin\theta$ < torque

质点的角动量定理: $\frac{d\vec{L}}{dt} = \frac{d(\vec{r}\times m\vec{v})}{dt} = \underbrace{\frac{d\vec{r}}{dt}}_{=\vec{v}}\times m\vec{v} + \vec{r}\times\frac{dm\vec{v}}{dt} = \vec{r}\times\vec{F} = \vec{M}$ 力矩 $\vec{M} = \vec{r}\times\vec{F} = d\vec{L}/dt = I\alpha$

$\int\vec{M}\,dt = \int d\vec{L} = \vec{L} - \vec{L_0}$

角动量的变化量等于τ关于t积分 ☆

moment of inertia & angular momentum

< particle $\vec{L} = r\times mwr = (mr^2)w_z$

rigid body $L_z = (\Sigma mr^2)w_z = Iw_z$

$\tau_z = \frac{dL_z}{dt} = \frac{d(Iw_z)}{dt} = I(\frac{dw_z}{dt}) = I\alpha_z$

rotational kinetic energy & torque

$K = \frac{1}{2}Iw^2 = \frac{1}{2}Iw_z^2$

power:

$P = \frac{dK}{dt} = \frac{d(\frac{1}{2}Iw^2)}{dt} = \frac{I}{2}2w\frac{dw}{dt} = I\alpha_z w_z = \tau_z w_z$

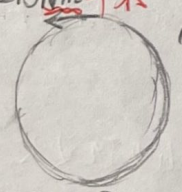
work done by torque = net torque × angular displacement along the same direction

$\int_{K_1}^{K_2} dK = \int_{t_1}^{t_2}\frac{dK}{dt}dt = \int_{t_1}^{t_2}\tau_z w_z dt = \int_{t_1}^{t_2}\tau_z(\frac{d\theta}{dt})dt$

$= \int_{\theta_1}^{\theta_2}\tau_z d\theta = W = K_2 - K_1 = \Delta K$

Problem 1

$\tau = 10Nm$ 牛米



shaft 轴

(i) $\tau_z = I\alpha_z$

$\therefore \alpha_z = 5 rad/s^2$

$I = 2kg\,m^2$

(ii) $t = 8s$

$\alpha_z = \frac{dw_z}{dt}$ $\int_0^8 5dt = w_f$

$\therefore w_f = 40 rad/s$

or $Iw_f = \int_0^8 \tau dt = 80$ $\therefore w_f = 40 rad/s$

$KE = \frac{1}{2}Iw^2 = \frac{1}{2}\cdot 2\cdot 1600 = 1600J = W$

(iii) $P = \frac{W}{t} = \frac{1600}{8} = 200W$

求合力矩时必须考虑(外)力的作用点(∵位置会影响力臂,进而影响力矩).

$\vec{M}_{外} = 0$, 若过程中M外恒为0, 则L守恒, $\frac{d\vec{L}}{dt} = 0$

< the conservation of angular momentum.

rotationally "completely inelastic collision"

$I_A w_A + I_B w_B = (I_A + I_B)w$

$\Rightarrow w = (I_A w_A + I_B w_B)/(I_A + I_B)$

Problem 2

(i) $w_f = \frac{I_1}{I_1+I_2}w_i$

(ii) $\frac{\frac{1}{2}(I_1+I_2)\frac{I_1^2}{(I_1+I_2)^2}w_i^2}{\frac{1}{2}I_1 w_i^2} = \frac{I_1}{I_1+I_2}$.

Problem 3

(i)

$r = 100m$

$m = 5\times10^4 kg$.

$w^2 r = g = 9.8$ $w = 0.313 rad/s$.

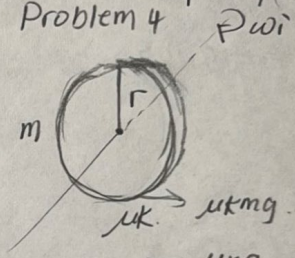
$\therefore L = \vec{r}\times\vec{p} = Iw = Mr^2 w = 1.57\times10^8 kgm^2/s$.

(ii) $L = \int_0^t \tau dt$ (angular impulse - angular momentum)

$\tau = r\times F = 2\times100\times1.5 = 2.5\times10^4 Nm$

$\therefore t = 1.57\times10^8/(6.5\times10^4) = 6.28\times10^3 s = 1.74h$

Problem 4



$Iw_i = \int \tau_{uk} mg dt$

$\therefore t = Iw_i/\mu_k mg$

$\therefore \Delta x = \frac{1}{2}\mu_k g\frac{I^2 w_i^2}{\mu_k^2 m^2 g^2}$

$= \frac{I^2 w_i^2}{2\mu_k m^2 g}$

$\rightarrow a = \mu_k g$

$I = \frac{1}{2}mr^2$ $\Delta x = \frac{m^2 r^4 w_i^2}{8\mu_k m^2 g} = \frac{w_i^2 r^4}{8\mu_k g}$ ✗

不要老是把I当未知量用

什么时候会只转动? 未考虑此状态.

$wr = v$ (纯滚动条件)

纯滚动不是纯转动, 而是转动+平动

$I(w_i - w_f) = \int \tau_{uk} mg dt$

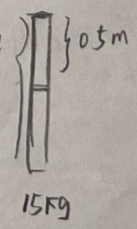
$\therefore t = I(w_i - w_f)/\mu_k mgr$

$v = \mu_k g t = I(w_i - w_f)/mr = w_f\cdot r$

$\frac{1}{2}mr^2(w_i - w_f) = w_f\cdot r^2$ $\therefore w_f = \frac{1}{3}w_i$

$\therefore \frac{1}{2}\mu_k g\left(\frac{1}{2}mr^2\cdot\frac{2}{3}w_i\right)^2 / \mu_k^2 m^2 g^2 r^2 = \frac{1}{18}\frac{w_i^2 r^2}{\mu_k g}$ ✓

Problem 5



$10^{-2}kg$ = 10g

1m, 0.5m, 400m/s, 15kg

$L_i = Iw$

$= mr^2 w = 10^{-2}\cdot(0.5)^2\cdot\frac{400}{0.5} = 2$

$= (mr^2 + \frac{1}{3}ML^2)w$ $w = 0.4 rad/s$

$10^{-2}\cdot0.5^2 + \frac{1}{3}15\cdot1$

另解: 取地面上B一点, ∵f径过此点, τ为0. angular momentum is conserved

$Iw_i = (I + mr^2)w_f$

$\therefore w_f = \frac{1}{3}w_i$. $v_f = r\cdot w_f = r(\frac{w_i}{3})$

$f\times\Delta t = mv_f - mv_i$

$\Delta x = \frac{1}{2}(v_i + v_f)\Delta t = \frac{r^2 w_i^2}{18\mu_k g}$

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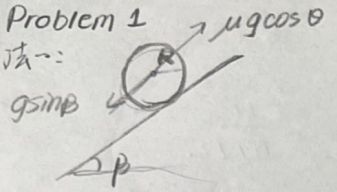
Dynamics of rolling motion

rolling rigid body < translational motion (cm) / rotation about an axis through cm

$KE = \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2$

Problem 1

法一:



$I = \frac{2}{5}mR^2$

$a = \frac{mg\sin\beta - \mu mg\cos\theta}{m}$

$\tau = I\alpha = \frac{2}{5}mR^2\alpha = \frac{\mu mg\cos\theta}{R} = \frac{F}{r}$

$\alpha = \frac{\mu g\cos\beta}{\frac{2}{5}R^2}$ 力矩是 $\vec{r}\times\vec{F}$, 不是 $\vec{F}$!!!

$\therefore g\sin\beta - \mu g\cos\beta = \frac{5}{2R}\mu g\cos\beta$

$g\sin\beta = \frac{7}{2}\mu g\cos\beta \quad \therefore \mu = \frac{2}{7}\tan\beta$

$\therefore a = g\sin\theta - \frac{2}{7}\tan\beta\, g\cos\beta = \frac{5}{7}g\sin\theta$

$f = \frac{2}{7}Mg\sin\beta$

纯滚动的核心运动条件是:

质心线速度 $v_{cm} = \omega R$

对t求导 $a_{cm} = \alpha_{cm}R$

法2. $Mgh = \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2$

$v_{cm} = R\omega$

$v_f^2 = 2a\Delta s = 2a\frac{h}{\sin\beta}$

$a = \frac{5}{7}g\sin\beta$

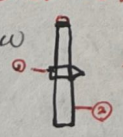
$Mg\sin\beta - f_s = Ma_{cm}$

$f_s = \frac{2}{7}Mg\sin\beta$

Problem 2 $(0.01\times0.5^2 + \frac{1}{3}\times15\times(2\times0.5)^2)\omega f$

$10^{-2}\times0.5^2\times\frac{+00}{0.5} = (10^{-2}+15)\cdot\frac{1}{3}0.2\,\omega$

??



分别算 $I\omega$

分成两部分看

$\omega = 0.4\ rad/s$

$\therefore KE = \frac{1}{2}(10^{-2}\times0.05^2 + \frac{1}{3}15(2\times0.05)^2)0.4^2$

$= (10^{-2}+15)g\Delta h_{cm}$

$\therefore \Delta h_{cm} = 2.72\times10^{-3}m$

Problem 3

$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

$\omega R = v \quad I = kmR^2$

$mgh = \frac{1}{2}mv^2 + \frac{1}{2}kmR^2\frac{v^2}{R^2}$

$v^2 = \frac{2gh}{1+k}$

Problem 4

$k = \frac{2}{5} \quad v^2 = \frac{2gh}{7/5} = \frac{10gh}{7}$

$\therefore v = \sqrt{\frac{10gh}{7}}$

$\therefore (0 + \sqrt{\frac{10gh}{7}})\cdot\frac{1}{2}\cdot t = \frac{h}{\sin\theta} \quad t = \frac{2h}{\sin\theta}\sqrt{\frac{7}{10gh}}$

☆a仅与g和θ有关, 静f不消耗能量, 只是用来提供转动的力矩(没有哪一处是有相对滑动的)

< translation $\sum F_i = ma$ ☆仅针对不打滑
rotation $\sum \tau_i = I\alpha$

做刚体的题不要老是把f忘了

Problem 5

(i) $F - \mu Mg = Ma$

$R: \mu Mg = I\alpha$
$a = \alpha R$
$\Big\}\ \mu Mg = \frac{I}{R}\frac{a}{R} = \frac{1}{2}Ma$

$a = \frac{2F}{3M}$

$\mu Mg = \frac{1}{2}M\frac{2F}{3M} \quad \mu = \frac{F}{3Mg}$

☆ 1. 只说固定轴转 $\frac{1}{2}I\omega^2$
整体在移动但不转 $\frac{1}{2}mv^2$
又移动又转(滚动) 都有

2. frictionless pulley 是说滑轮的轴(中心无f)而不是说绳与pulley间无f

3. 绳子和滑轮间有f, 这时滑轮两边张力可以不同 "流程/pulley"

4. 一直 rolls without slipping, $v = \omega R$. 即v与ω同时为0. 即 $\frac{1}{2}mv^2$ & $\frac{1}{2}I\omega^2$ 同时为0

5. 轻绳m=0, 张力处处相等

6. I: $kg\cdot m^2 = \int mr^2$
α: rad/s^2
τ: $N\cdot m = \vec{r}\times\vec{F} = d\vec{L}/dt = I\alpha$
L: $kg\cdot m^2/s = r\times p = I\omega$

7. 对于绳子假设在物体上这类model, 默认接触处不滑动, 因此必有 $v = \omega R$.

8. 能量关回算 $\frac{1}{2}I\omega^2$ 它配完不一样

9. 看滑轮往哪个方向转, 哪一侧张力就更大

T_1 f ← T_2 → 取出 T_1 ← f → T_2 (m≈0) $\therefore T_2 = T_1 + f$

角冲量=角动量的变化 $\vec{r}\times\vec{J} = \Delta\vec{L}$

若物体初始不转: $rJ = I\omega$, $\omega = \frac{rJ}{I}$

①平动(所有点一样): v_{cm} 整个物体像整体平移

②转动(每个点不同): ω×r 离质心越远转得越快

☆补充: 记: 对任意参考点(轴), 刚体总角动量可写成

$L_O = r_{cm}\times mv_{cm} + I_{cm}\omega$ 前提是说刚体的质心转动

$= \sum \vec{r_i}\times m_i\vec{v_i} = \sum(\vec{r_{cm}} + \vec{r_i}')\times m_i(\vec{v_{cm}} + \vec{v_i}')$ $\times m_i\vec{v_i}'$

$= \underbrace{\sum \vec{r_{cm}}\times m_i\vec{v_{cm}}}_{r_{cm}\times mv_{cm}} + \underbrace{\sum \vec{r_{cm}}\times m_i\vec{v_i}'}_{①=0} + \underbrace{\sum \vec{r_i}'\times m_i\vec{v_{cm}}}_{②=0} + \underbrace{\sum \vec{r_i}'}_{I_{cm}\omega}$

①.②: 从①的eq. ③同理

$\Rightarrow \vec{r_{cm}}\times\underbrace{\sum m_i\vec{v_i}'}_{=0} \quad \because \vec{v_i}' = \vec{v_i} - \vec{v_{cm}}$

$\therefore \sum m_i(\vec{v_i} - \vec{v_{cm}}) = \sum m_i\vec{v_i} - \sum m_i\vec{v_{cm}} = M\vec{v_{cm}} - M\vec{v_{cm}} = 0$

∴在质心参照系中: 所有质点的"质量加权平均位置"和"质量加权平均速度为0". "质量加权平均加速度为0"

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